EVALUATION OF NLP SYSTEMS

Lecture 7 - Best Practices

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- some issues to be aware of

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Putting the Linguistics in Computational Linguistics

Emily Bender (2017)

not all papers need to be linguistically informative—
but they should all be linguistically informed

1. Know your data

- look at training data
- what are its properties?
- language varieties represented
- long and complex or short and simple utterances?
- any repeated items? why?
- well-edited or messy?
- any incompatibility with pretrained components?

2. Describe your data honestly

- describe variety, origin, genre
- English is not 'Language', it's a language
- English is not a canonical human language!
- Bender rule: say which language, also when it's English

3. Focus on linguistic structure

- don't look past linguistic structures
- communicate and discuss with linguists

4. Do (linguistic) error analysis

- collect a sample (or all) errors in the dev set
- classify them
- what recurring linguistic patterns result in errors?
- good deal of linguistic insight
- colaborate with linguists

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Improving Replication and Reproduction in Computational Linguistics

Antske Fokkens (2018)

replication

obtaining the same results using the same experiment

sharing code and resources whenever possible is important:

- openly verifiable
- provides additional means for other researchers to build their work directly upon previous work (rather than first going through the frustration of reimplementing a new baseline system good enough to test their hypotheses on)

reproduction

reaching the same conclusion through different means

- beating SOTA should not be the ultimate goal
- evaluation should provide insight into how a model works, what phenomena it captures or which patterns the machine learning algorithm picked up, compared to alternative approaches
- papers should be evaluated based on what they teach us, i.e. whether they verify their hypothesis by comparing it to a suitable baseline
- report negative results to avoid that other people waste time on something you know doesn't work

Example of reproducibility checklist

For all reported experimental results:

- [ ] A clear description of the mathematical setting, algorithm, and/or model.
- [ ] A link to a downloadable source code, with specification of all dependencies, including external libraries
- [ ] Description of computing infrastructure used
- [ ] Average runtime for each approach
- [ ] Number of parameters in each model
- [ ] Corresponding validation performance for each reported test result
- [ ] Explanation of evaluation metrics used, with links to code

For all experiments with hyperparameter search:

- [ ] Bounds for each hyperparameter
- [ ] Hyperparameter configurations for best-performing models
- [ ] Number of hyperparameter search trials
- [ ] The method of choosing hyperparameter values (e.g., uniform sampling, manual tuning, etc.) and the criterion used to select among them (e.g., accuracy)
- [ ] Expected validation performance, as introduced in Section 3.1 in * Dodge et al, 2019, or another measure of the mean and variance as a function of the number of hyperparameter trials.

For all datasets used:

- [ ] Relevant statistics such as number of examples
- [ ] Details of train/validation/test splits
- [ ] Explanation of any data that were excluded, and all pre-processing steps
- [ ] A link to a downloadable version of the data
- [ ] For new data collected, a complete description of the data collection process, such as instructions to annotators and methods for quality control.

EMNLP 2020

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The field of natural language processing is chasing the wrong goal

Jesse Dunietz (2020)

what has the world really gained if a massive neural network achieves SOTA on some benchmark by a point or two?

winning the leaderboard is an exercise that may not make real-world tools any better

in the quest to reach difficult benchmarks, evaluations have lost sight of the real targets: sophisticated downstream applications

many apparent improvements emerge not from general comprehension abilities, but from models' extraordinary skill at exploiting spurious patterns

however systems are implemented, if they need to have faithful world models, then evaluations should systematically test whether they have faithful world models

The Social Impact of Natural Language Processing

considering the ethical implications of NLP

Hovy & Spruit (2016)

data sciences now affect human subjects in real time (e.g. social networks)

the texts we use in NLP carry latent information about the author and situation

language is a proxy for human behavior, and a strong signal of individual characteristics

what research is carried out, and its quality, directly affect the functionality and impact of those technologies

the mutual relationships between language, society, and the individual are the source for the societal impact factors of NLP

> exclusion or misrepresentation due to demographic bias
side-effect of data

> topic overexposure / availability heuristic
> underexposure
side-effects of research design

> bias confirmation

> overgeneralization
side-effect of modeling

> dual use

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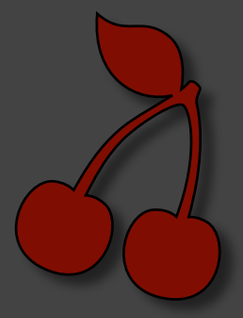
Use Datasets Wisely

Choose appropriate datasets to evaluate on, not only those that 'everyone uses'.

A dataset is a sample. It's neither 'Natural Language' nor the whole population.

Extrapolation and inference must be done with caution!



don't cherry-pick only good results

We need unseen data to evaluate our models. Use the test set properly: Validate only on the validation set, test only once in the test set.

Even so, a whole community using the same test set 'only once' over and over again can also be problematic ('group overfitting').

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Caring About Visualization

design nice plots and tables

make your data and metrics visually understandable

communicate your message with images

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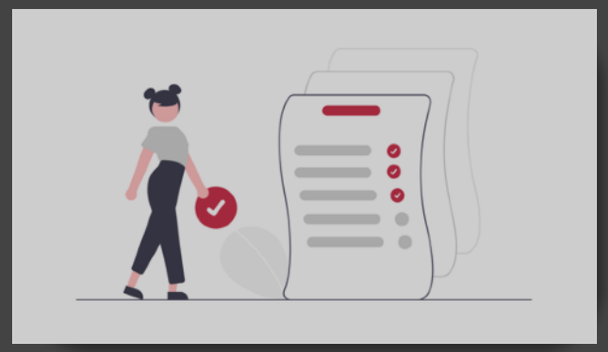
A Few Points to Keep in Mind



- rigour and formality in the evaluation process are essential
- think critically about every decision
- report results correctly, methodically and clearly
- be honest
- acknowledge limitations
- don't oversell

- mind the Linguistics (it's NLP for a reason)
- beating SOTA and chasing leaderboards *per se* should not be our main purpose
- science is cooperation, not competition
- ethics: think, read, discuss about it
- replication & reproduction: think of others who want to test and use your system/code

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See you next year!

- ☑ some issues to be aware of

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